

0.1 Probability Spaces

Definition 0.1.1 ► Probability Spaces

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \rightarrow [0, 1]$$

Example 0.1.2 ► Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

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